Australian Overseas Migration — End-to-End Analytics in Databricks

> In 2020-21, Australia's net overseas migration went **negative** for the first time in decades — more people left than arrived. A year later it hit a record high. This project traces the whole 21-year story.

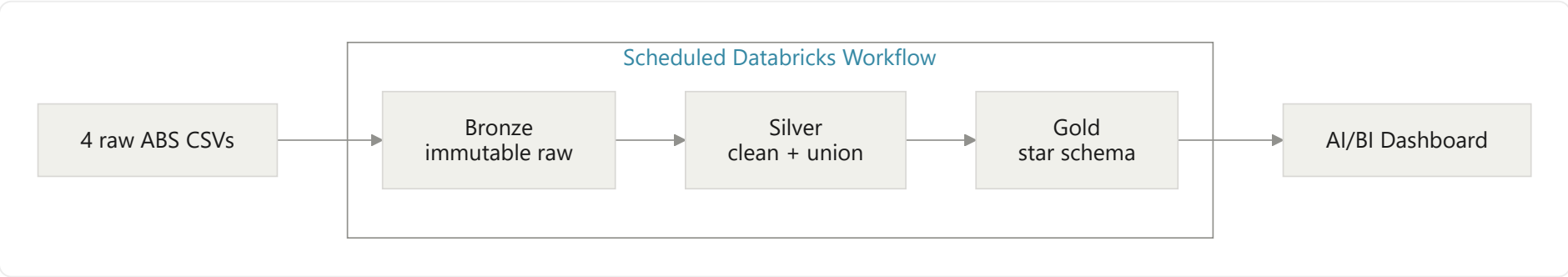
An end-to-end analytics pipeline built on 21 years of Australian Bureau of Statistics (ABS) migration data (2004-05 to 2024-25). Four raw ABS Excel files are cleaned, reshaped, moved through a bronze/silver/gold medallion architecture in Databricks, modelled as a star schema, and surfaced in a 3-page AI/BI dashboard.

Key findings

- **Net migration reversed in 2020-21.** It fell to about **−85,000** when COVID closed the borders — the first negative year in decades — then rebounded to a record **~538,000** in 2022-23.
- **Temporary visas drive migration, not permanent ones.** Temporary visas added ~1.7M net over the period; permanent visas ~497K. Most permanent residents arrive on a temporary visa first and convert onshore.
- **India and China lead source countries.** India tops net migration (~867K over 21 years) and overtook China around 2018-19. In 2024-25, India (~74.5K) and China (~69.9K) were the two largest arrival sources.
- **Migration concentrates in two states.** New South Wales and Victoria receive the bulk of arrivals, in line with population.

Architecture

Raw files are uploaded once; the three middle layers run as a scheduled job; the dashboard reads the gold tables.



- **Bronze** — raw CSVs loaded as-is into Delta tables with lineage columns (ingest time, source file). No cleaning.

- **Silver** — the three measure tables (net, arrivals, departures) unioned into one, types fixed, quality flags added (`is_preliminary`, `is_national_total`). Visa file cleaned separately.
- **Gold** — a star schema: one fact table (`fact_migration`) and three dimensions (`dim_country`, `dim_geography`, `dim_year`).

The dashboard — 3 pages

Page 1 — National Migration Overview

The national picture across all 21 years.

- **Total arrivals by state/territory** (bar) — NSW (3.44M) and Victoria (2.73M) dominate, down to Tasmania (93.3K). National total excluded to avoid double-counting.
- **Arrivals, departures and net by year** (line) — the headline chart. Arrivals and departures normally run far apart; in 2020-21 arrivals collapse and net dips below zero; 2022-23 rebounds to a record.
- **Arrivals vs departures by visa group** (grouped bar) — temporary visas carry the most movement; Australian citizens arrive and depart in near-equal numbers.

Page 2 — Net Migration (the bottom line)

Focuses on net migration (arrivals minus departures), the true measure of population gain.

- **Net migration by visa group** (bar) — temporary visas ~1.7M, permanent ~497K, New Zealand citizens ~430K, and **Australian citizens slightly negative (~−243K)** — more leave than return.
- **Top 10 countries' net migration by year** (multi-line) — shows India overtaking China after 2018-19, and the shared 2020-21 dip (China falls hardest, to ~−50K).
- **Net migration by financial year** (area) — the same reversal-and-rebound shape as the national net line, emphasising the 2020-21 trough and 2022-23 peak.

Page 3 — Source Countries & Latest Year

Breaks the most recent year (2024-25) down by origin and destination.

- **Arrivals vs departures by state** (combo: bars + line) — NSW and Victoria lead on both; departures follow the same order at a lower level.
- **Arrivals by source country** (choropleth map) — arrivals concentrate in South and East Asia, with India and China far ahead. Uses a single-hue sequential colour scale (darker = more).

Data quality

Real issues found and fixed — the part of the project that matters most:

- **Net = arrivals − departures reconciliation.** Confirmed the three measures are internally consistent within ABS rounding (nearest 10).
- **Geography double-count.** "Australia" in the data is the national total **and** equals the eight states summed. Charts that forgot a `geography = 'Australia'` filter doubled every national figure (net showed ~1.08M instead of the true ~538K). Fixed by filtering geography explicitly.
- **The country-vs-geography trap.** "Australia" appears as both a country of birth (Australian-born) and a geography. The `is_national_total` flag isolates that row so country rankings don't double-count it.
- **Subtotal rows.** The visa table contains both detail rows and subtotals ("Total", "Total permanent visas"). Excluded via `is_subtotal = false` so totals aren't inflated.

Tech stack

- **Databricks** (Free Edition) — serverless compute, Delta Lake, AI/BI dashboards
- **PySpark** — ingestion and transformation
- **SQL** — modelling and analysis
- **Source:** Australian Bureau of Statistics, Overseas Migration release

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Repository structure

```
notebooks/
  01_bronze.py        # raw ingestion
  02_silver.py        # clean, union, quality flags
  03_gold.py          # star schema (fact + dimensions)
  04_analysis.sql     # analysis queries
data/
  *_tidy.csv          # cleaned, upload-ready files
README.md
```

How to run

1. Create a free Databricks Free Edition workspace (not Community Edition — retired Jan 2026).
2. Create a schema: `CREATE SCHEMA IF NOT EXISTS workspace.migration;`
3. Upload the four `*_tidy.csv` files to a Volume.
4. Import and run the notebooks in order: `01_bronze` → `02_silver` → `03_gold`.
5. Run `04_analysis.sql` in the SQL editor, then build the dashboard on the gold tables.
6. (Optional) Chain the three notebooks into a scheduled Workflow with `01 → 02 → 03` dependencies.

Scope (honest note)

This is a learning project built to practise the Databricks platform and the medallion/dimensional-modelling method on realistic, messy government data. The dataset is ~142,000 rows — small by design. It demonstrates ingestion, layered architecture, modelling, data-quality rigour, and BI delivery; it is **not** a big-data or real-time system.

Screenshots

Add your dashboard screenshots here:

Page 1 — National Overview Page 2 — Net Migration Page 3 — Source Countries